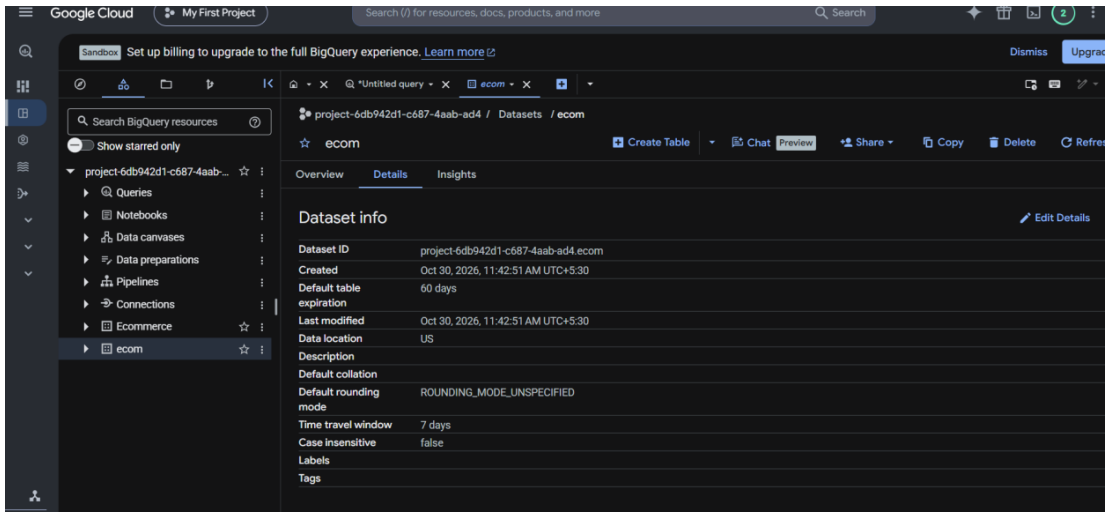
HCL TECH
TRAINING PRACTICE REPORT

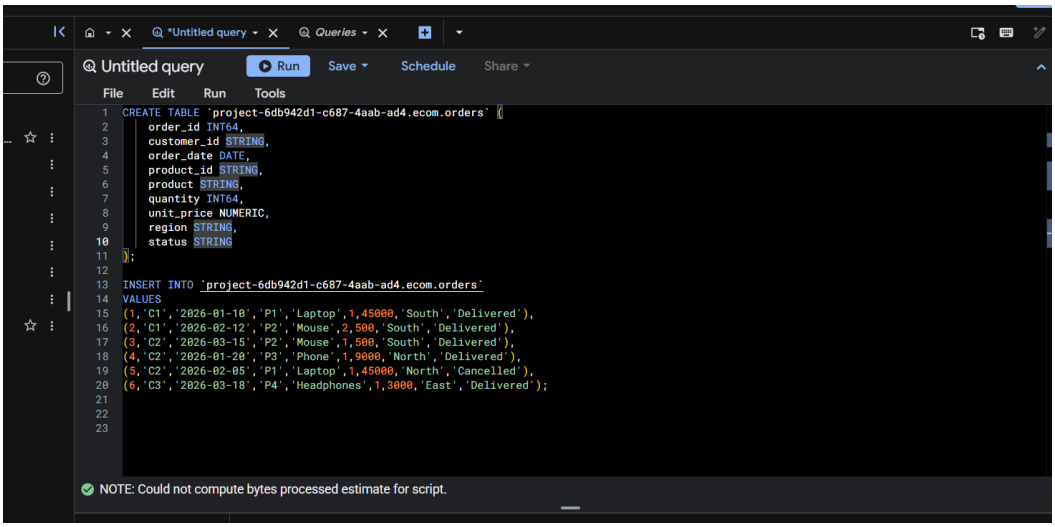
E-Commerce Sales Analysis

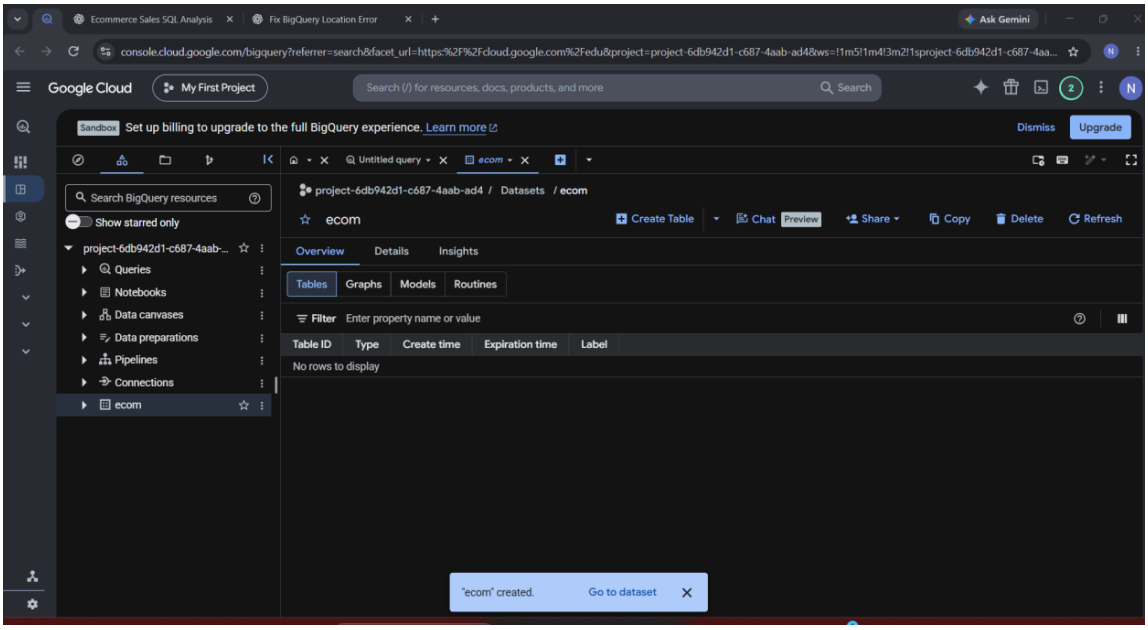
DETAIL	INFORMATION
Prepared by	Harshini B
Registration Number	23BTRIS014
HCL ID	HCLTFP2260284
Programme	B.Tech – Information Science and Engineering
Institution	Jain (Deemed-to-be University), Bengaluru
Platform	Google Cloud Platform (GCP) – BigQuery
Dataset	ecom.orders
Date	30 October 2026

A eccomerce company receives 1000 orders daily. The business team wants to identify its highest revenue sales and top customer and monthly sale

Environment Setup and Table Creation



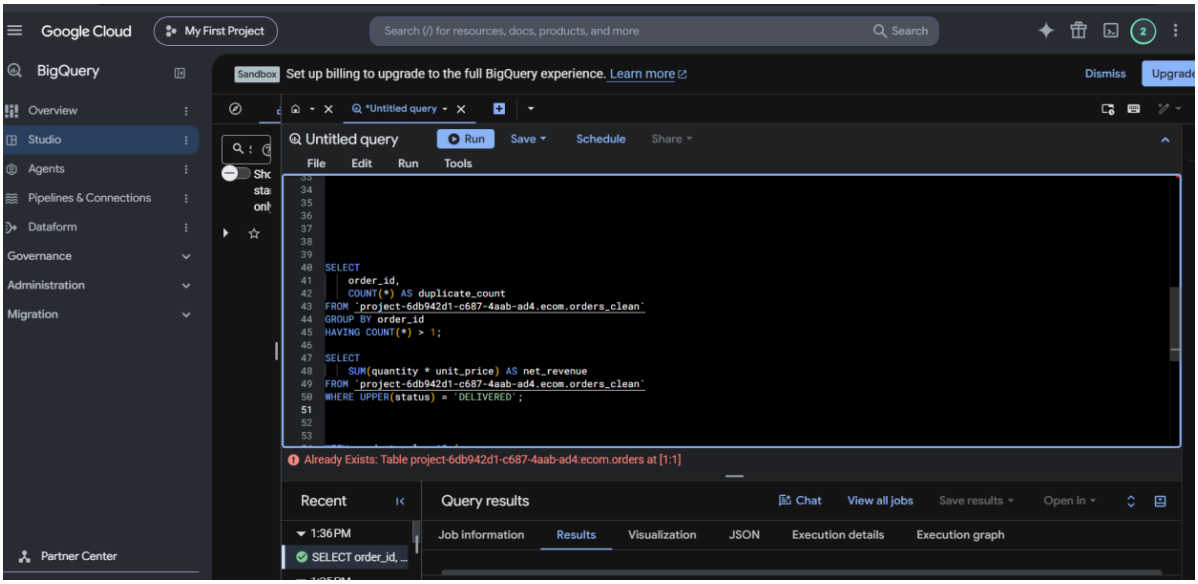




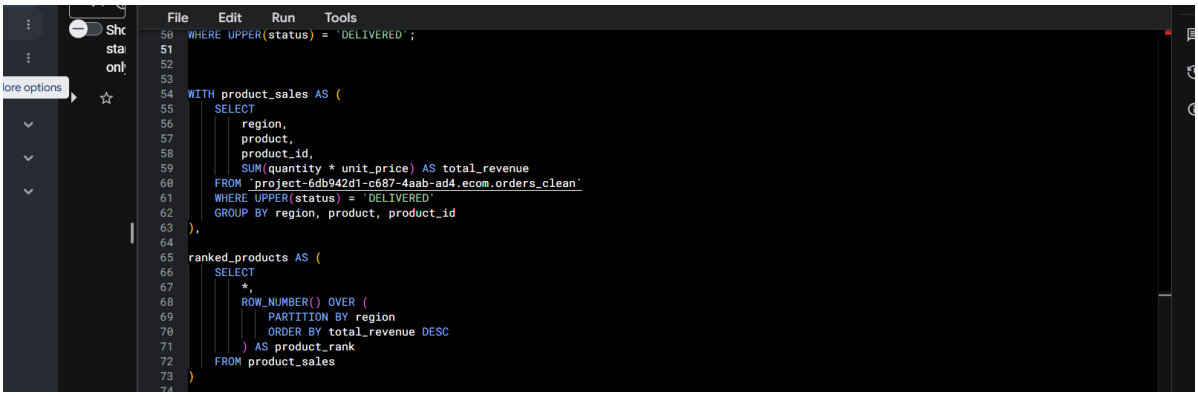
Task 1: Remove duplicate orders

```sql
CREATE OR REPLACE TABLE ecom.orders_clean AS
SELECT * EXCEPT (rn)
FROM (
 SELECT *,
 ROW_NUMBER() OVER (PARTITION BY order_id ORDER BY order_date DESC) AS rn
 FROM ecom.orders
)
WHERE rn = 1;
```

Task 2: Net revenue



Task 3: Top 3 products in each region





Task 4: Identify inactive customers

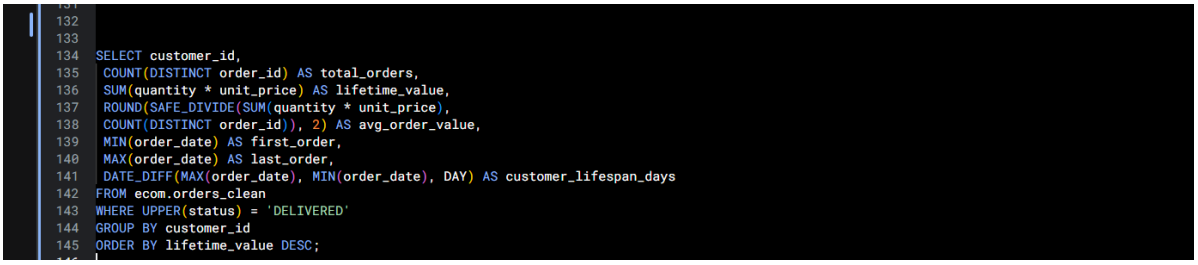


```
    WHERE UPPER(status) = 'DELIVERED'
    GROUP BY region, product, product_id
),

ranked_products AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY region
            ORDER BY total_revenue DESC
        ) AS product_rank
    FROM product_sales
)

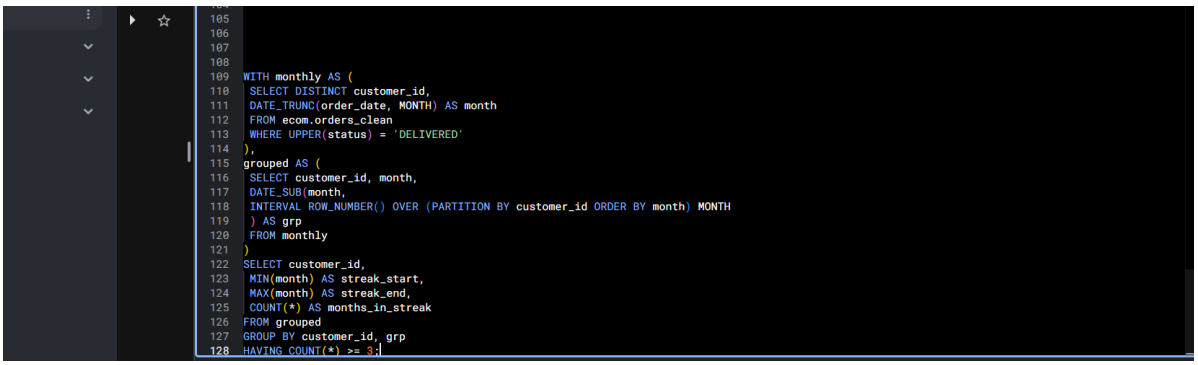
SELECT *
FROM ranked_products
WHERE product_rank <= 3
ORDER BY region, product_rank;
```

Task 6: Customer lifetime value (CLV) report



```
SELECT customer_id,
 COUNT(DISTINCT order_id) AS total_orders,
 SUM(quantity * unit_price) AS lifetime_value,
 ROUND(SAFE_DIVIDE(SUM(quantity * unit_price),
 COUNT(DISTINCT order_id)), 2) AS avg_order_value,
 MIN(order_date) AS first_order,
 MAX(order_date) AS last_order,
 DATE_DIFF(MAX(order_date), MIN(order_date), DAY) AS customer_lifespan_days
FROM ecom.orders_clean
WHERE UPPER(status) = 'DELIVERED'
GROUP BY customer_id
ORDER BY lifetime_value DESC;
```

Task 5: Customers who purchased for 3 continuous months



```
WITH monthly AS (
 SELECT DISTINCT customer_id,
 DATE_TRUNC(order_date, MONTH) AS month
 FROM ecom.orders_clean
 WHERE UPPER(status) = 'DELIVERED'
),
grouped AS (
 SELECT customer_id, month,
 DATE_SUB(month,
 INTERVAL ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY month) MONTH
 ) AS grp
 FROM monthly
)
SELECT customer_id,
 MIN(month) AS streak_start,
 MAX(month) AS streak_end,
 COUNT(*) AS months_in_streak
FROM grouped
GROUP BY customer_id, grp
HAVING COUNT(*) >= 3;
```

Task 7: Monthly sales

QUE RY

SELECT DATE_TRUNC(order_date, MONTH) AS sales_month,

COUNT(DISTINCT order_id) AS total_orders,

SUM(quantity * unit_price) AS monthly_sales

FROM ecom.orders_clean WHERE UPPER(status) = 'DELIVERED'

GROUP BY sales_month ORDER BY sales_month;

Task 8: Unit price below 10,000 in a particular month

QUE RY

SELECT order_id, customer_id, order_date, product, unit_price, quantity

FROM ecom.orders_clean

WHERE unit_price < 10000 AND order_date BETWEEN '2026-03-01' AND '2026-03-31' ORDER BY unit_price DESC;

Task 9: Customers who bought a product multiple times in 6 months

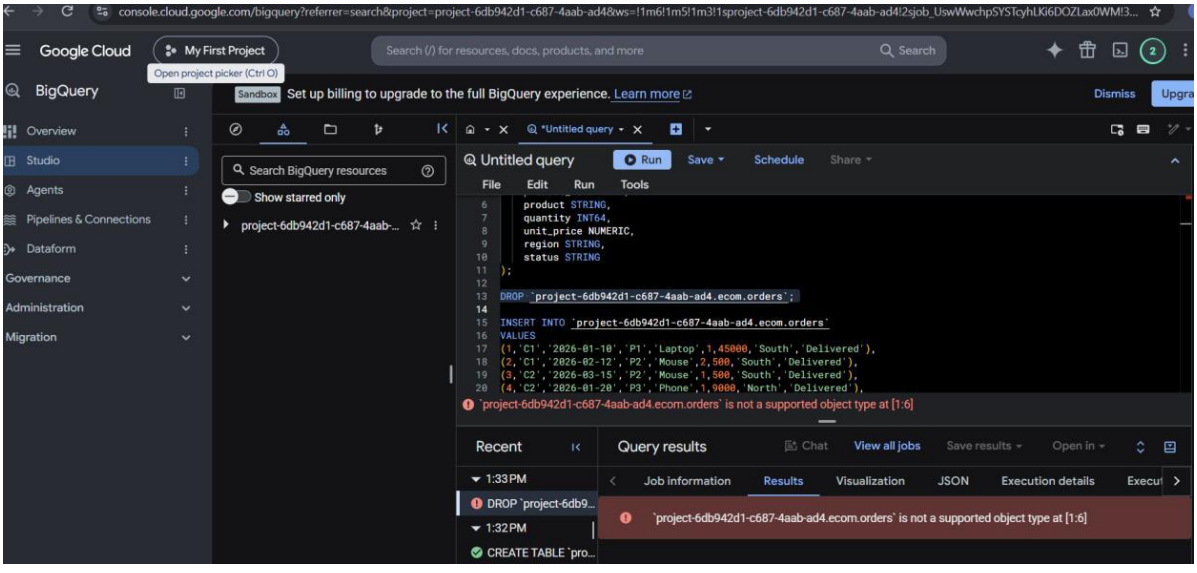
QUE RY

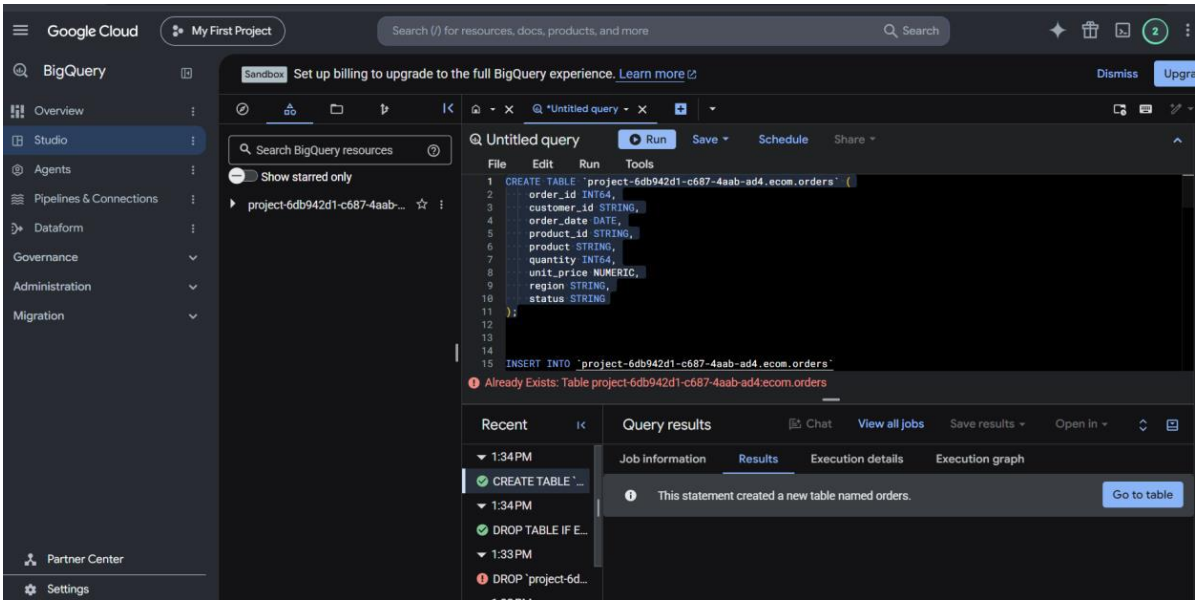
```
SELECT customer_id, product_id, product,
        COUNT(*) AS times_purchased,
        SUM(quantity) AS total_units
FROM ecom.orders_clean
WHERE product_id = 'P2'
       AND order_date BETWEEN '2026-01-01'
       AND '2026-06-30' AND UPPER(status) = 'DELIVERED'
GROUP BY customer_id, product_id, product HAVING COUNT(*) > 1;
```

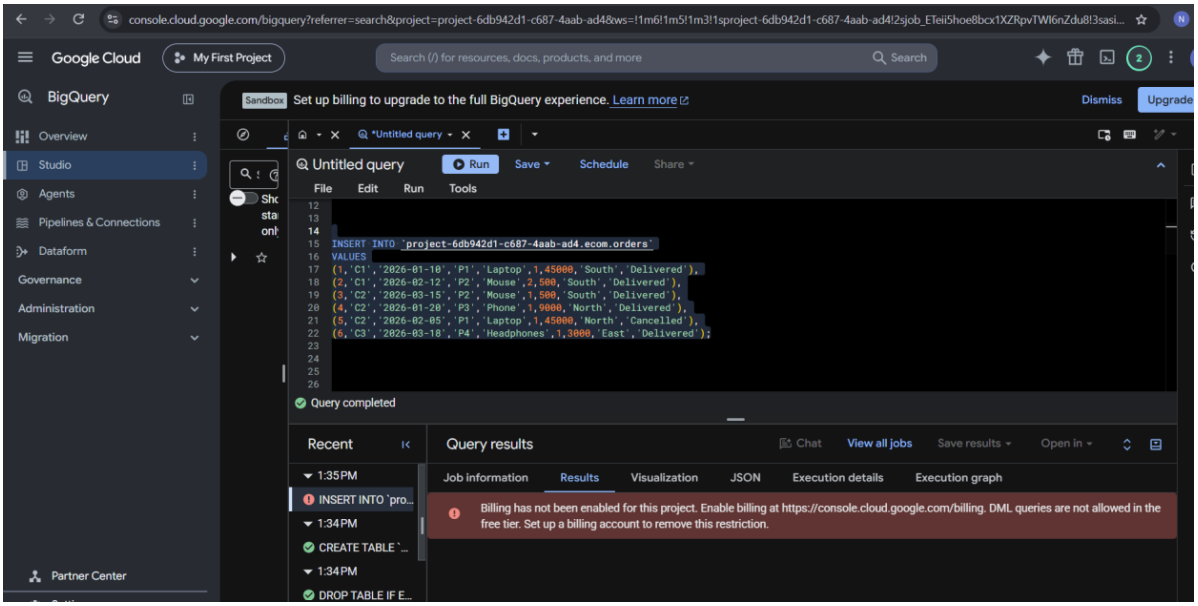
Task 10: Total sales for each product in a particular month

QUE RY

```
SELECT product_id, product,
SUM(quantity) AS units_sold,
SUM(quantity * unit_price) AS total_sales
FROM ecom.orders_clean WHERE UPPER(status) = 'DELIVERED'
        AND order_date BETWEEN '2026-01-01' AND '2026-01-31'
GROUP BY product_id, product ORDER BY total_sales DESC;
```




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project-6db942d1-c687-4aab-...
*Untitled query
Untitled query
Run
Save
Schedule
Share
File
Edit
Run
Tools
CREATE TABLE `project-6db942d1-c687-4aab-ad4.ecom.orders` (
order_id INT64,
customer_id STRING,
order_date DATE,
product_id STRING,
product STRING,
quantity INT64,
unit_price NUMERIC,
region STRING,
status STRING
);
INSERT INTO `project-6db942d1-c687-4aab-ad4.ecom.orders`
Already Exists: Table project-6db942d1-c687-4aab-ad4:ecom.orders
Recent
Query results
Chat
View all jobs
Save results
Open in
Job information
Results
Execution details
Execution graph
1:34 PM
CREATE TABLE `...
1:34 PM
DROP TABLE IF E...
1:33 PM
DROP `project-6d...
This statement created a new table named orders.
Go to table


console.cloud.google.com/bigquery?referrer=search&project=project-6db942d1-c687-4aab-ad4&ws=!1m6!1m5!1m3!1sproject-6db942d1-c687-4aab-ad4!2sjob_ETeii5hoe8bcx1XZRpvTWl6nZdu8!3sasi...
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Governance
Administration
Migration
Partner Center
*Untitled query
Untitled query
Run
Save
Schedule
Share
File
Edit
Run
Tools
INSERT INTO `project-6db942d1-c687-4aab-ad4.ecom.orders`
VALUES
(1,'C1','2026-01-10','P1','Laptop',1,45000,'South','Delivered'),
(2,'C1','2026-02-12','P2','Mouse',2,500,'South','Delivered'),
(3,'C2','2026-03-15','P2','Mouse',1,500,'South','Delivered'),
(4,'C2','2026-01-20','P3','Phone',1,9000,'North','Delivered'),
(5,'C2','2026-02-05','P1','Laptop',1,45000,'North','Cancelled'),
(6,'C3','2026-03-18','P4','Headphones',1,3000,'East','Delivered');
Query completed
Recent
Query results
Chat
View all jobs
Save results
Open in
Job information
Results
Visualization
JSON
Execution details
Execution graph
1:35 PM
INSERT INTO `pro...
1:34 PM
CREATE TABLE `...
1:34 PM
DROP TABLE IF E...
Billing has not been enabled for this project. Enable billing at https://console.cloud.google.com/billing. DML queries are not allowed in the free tier. Set up a billing account to remove this restriction.